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Summer Program

Identifying Aggressive Glioblastoma Profiles: A Composite Expression Multi-Gene Biomarker



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Problem Statement

- Most lethal primary brain tumor in adults.
- Tumor cells aggressively invade healthy brain tissue
- High cellular variance

Accurate Biomarkers are urgently needed to track disease progression and tumor borders



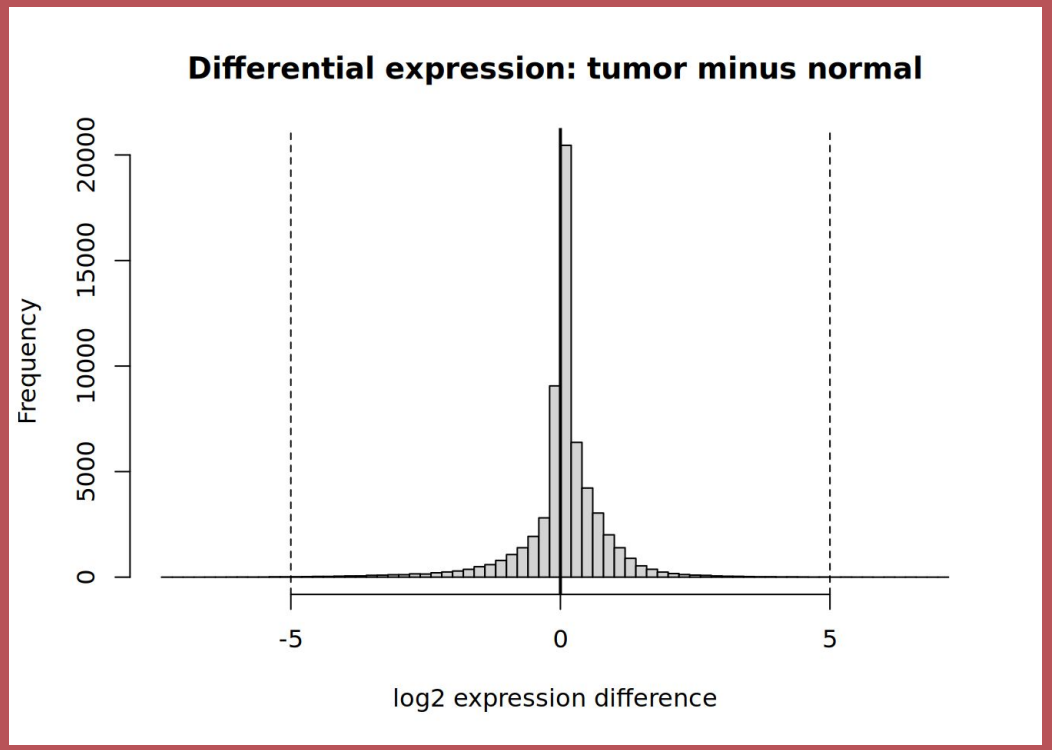
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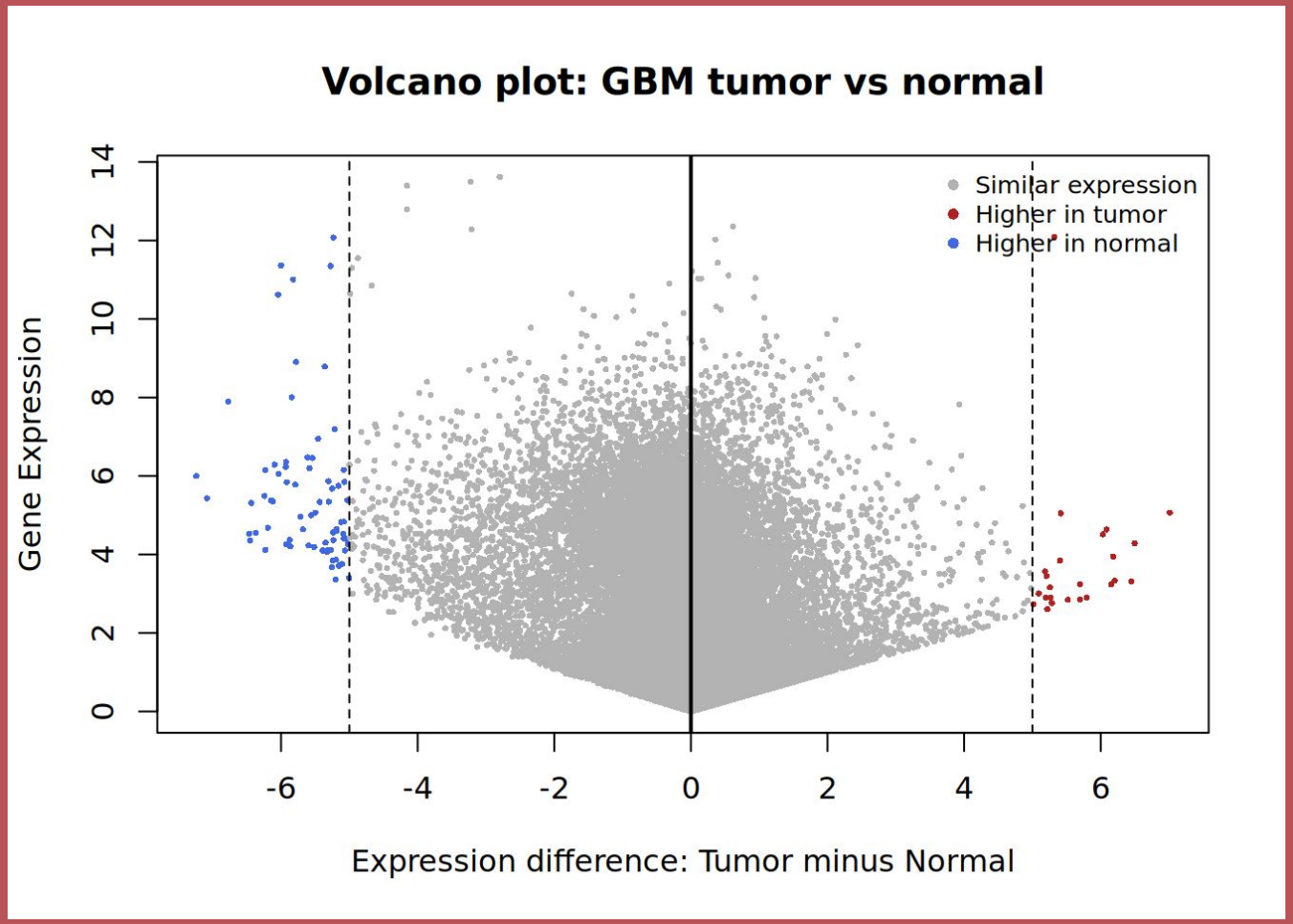
Methodology



- The Cancer Genome Atlas (TCGA) GBM dataset
- 391 Total Samples (372 Primary Tumors, 14 Recurrent Tumors, 5 Solid Normal Controls)
- 60,660 Genes, 391 Samples

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Selection Criteria: Top 15 Unregulated VS Downregulated genes

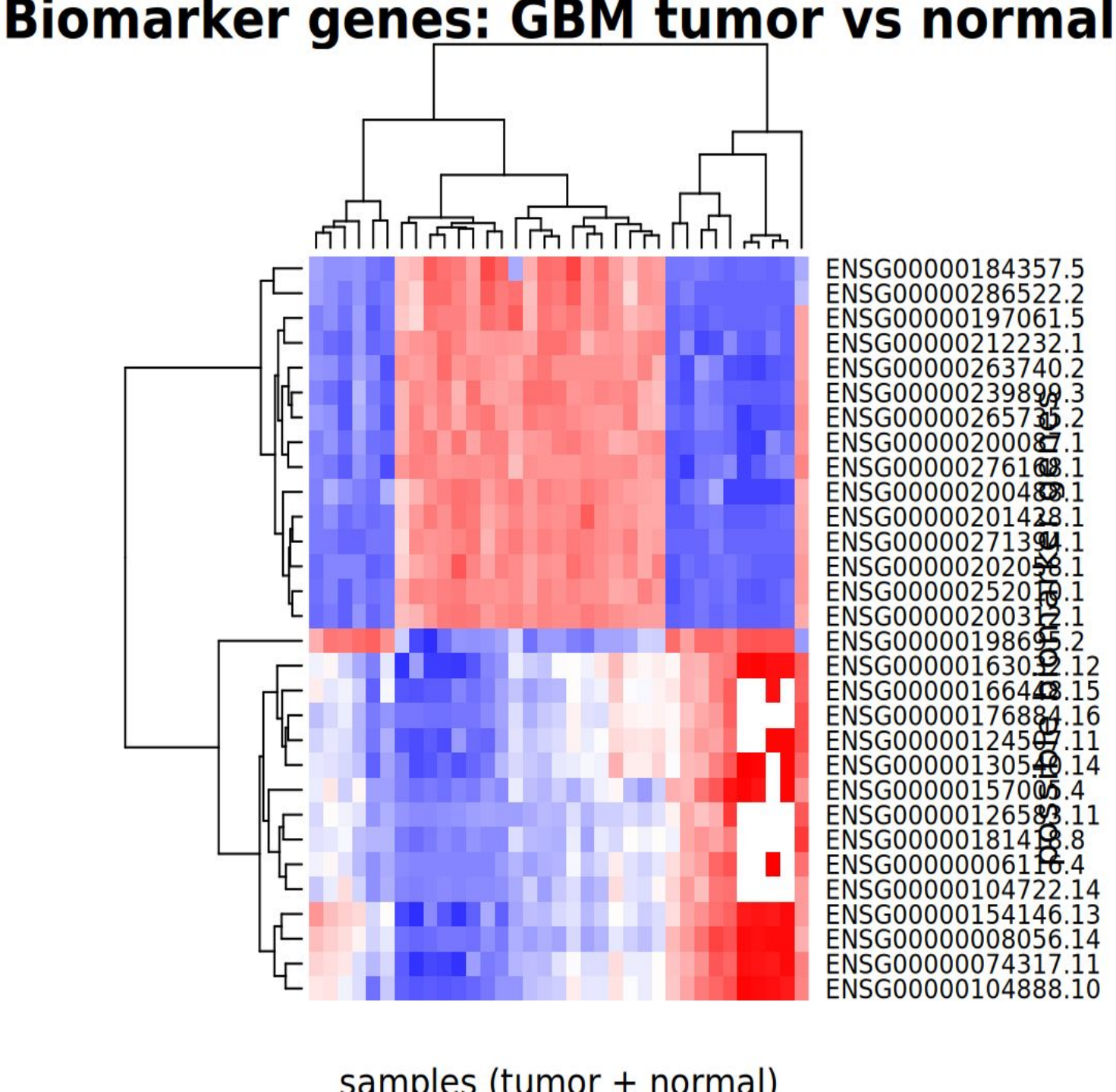
MALAT1: Upregulated

MT-ND6: Downregulated

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Gene Signature Visualization

Clear binary expression inversion

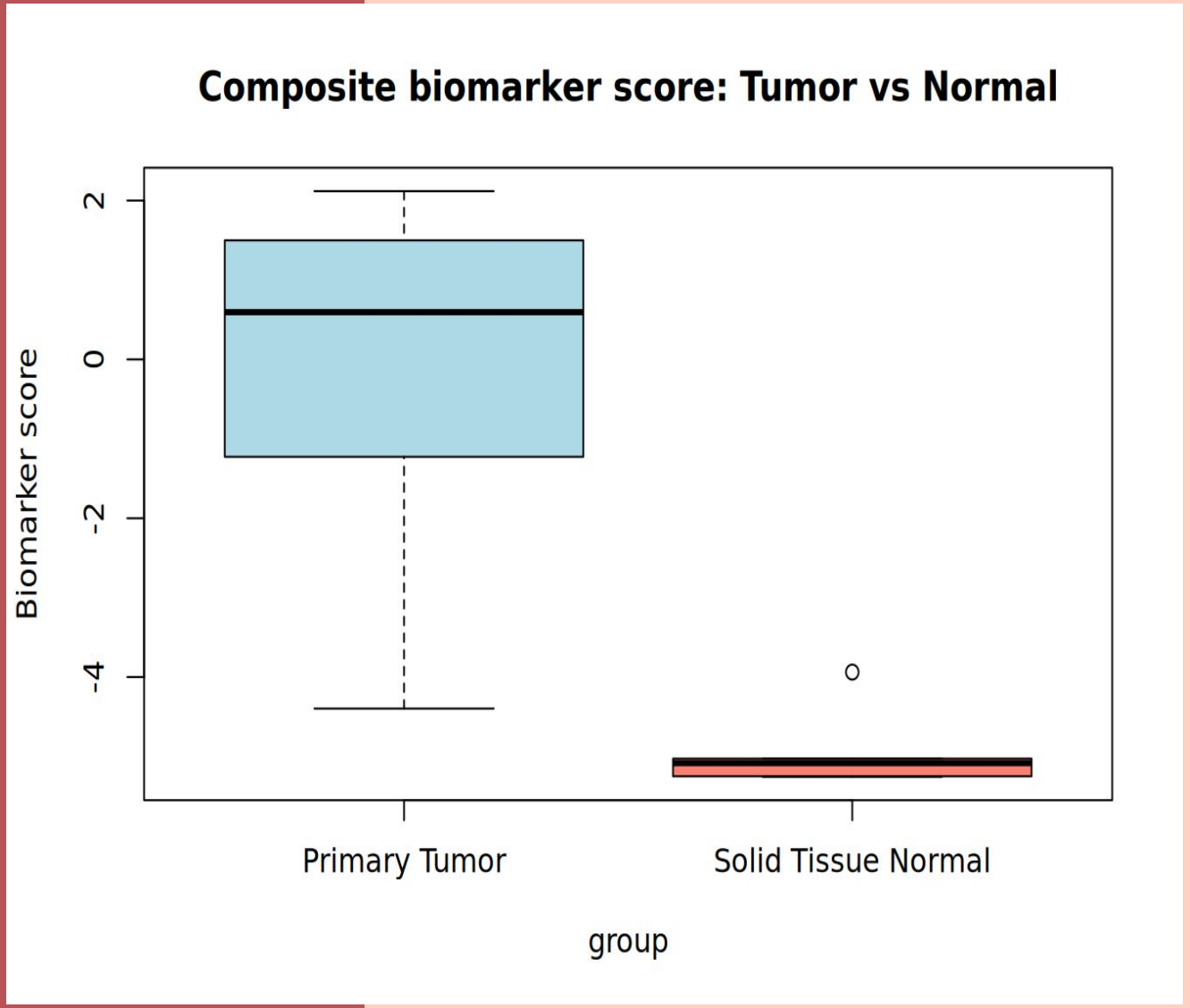


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Formulation/Validation

- Absolute separation of IQR ranges between groups

Statistically significant difference between both tumor groups; biomarker is viable



Implications & Limitations

- **Limitations:**
  - Highly unbalanced cohorts (372 Tumors vs. 5 Healthy Controls) from TCGA.
  - Validation is purely computational; lacks experimental bench verification.
- **Clinical Implications:**
  - Potential translation into blood or CSF-based **liquid biopsies** to avoid dangerous repeated brain biopsies.
  - Stratifying tumor aggression to personalize patient treatment plans.

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Acknowledgments/References

Systems Biology

DREAM-High Team

TCGA

https://portal.gdc.cancer.gov/projects/TCGA-GBM
https://rpubs.com/musfirah/1451746